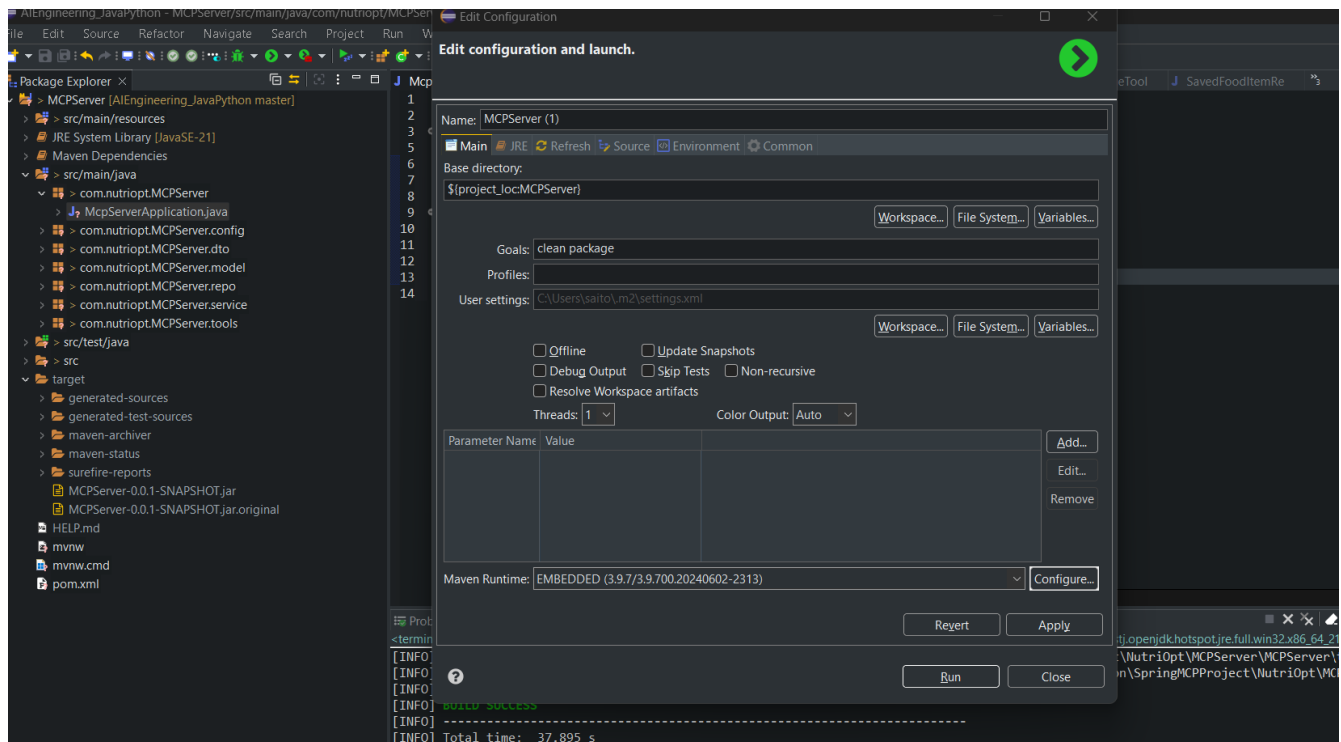


MCPServer Inspection



right-click main file and select Maven build

enter “clean package”

Look for “target/MCPServer-0.0.1-SNAPSHOT.jar”

Run “npx @modelcontextprotocol/inspector java -jar target\MCPServer-0.0.1-SNAPSHOT.jar”

Open <http://localhost:6274/?>

[MCP_PROXY_AUTH_TOKEN=a3d5532d36bdb32c8bccab9215e86d27551a9ff5dfef594b516c77f16da7bfef#tools](http://localhost:6274/?MCP_PROXY_AUTH_TOKEN=a3d5532d36bdb32c8bccab9215e86d27551a9ff5dfef594b516c77f16da7bfef#tools)

Click Connect

Transport Type

STDIO

Command

java

Arguments

-jar target/MCPServer-0.0.1-SNAPSH

> Environment Variables

Server Entry

Servers File

> Authentication

> Configuration

Connect

Disconnected

History

No history yet

Testing fresh-mcp-ping tool

MCP Inspector v0.22.0

Transport Type

STDIO

Command

java

Arguments

-jar target/FreshMcpPing-0.0.1-SNA

> Environment Variables

Server EntryServers File

> Authentication

> Configuration

RestartDisconnect

Connected

fresh-mcp-ping

Version: 0.0.1

Logging Level

debug

ResourcesPromptsToolsTasksAppsPingSamplingElicitationsRootsAuth

Tools

List Tools

Clear

ping
Simple ping test tool

Select a tool

Select a tool from

History

Clear

3. tools/list

2. logging/setLevel

1. initialize

Server Notificat

No notifications yet

ResourcesPromptsToolsTasksAppsPingSamplingElicitationsRootsAuthMetadata

Tools

List Tools

Clear

ping
Simple ping test tool

ping

Simple ping test tool

X Read-only✓ DestructiveX Idempotent✓ Open-world

Tool-specific Metadata:
No metadata pairs.

Run Tool

Copy Input

Tool Result: Success

""pong""

Now testing nutriopt-mcp-server with Ping tool, it works

MCP Inspector v0.22.0

Transport Type

STDIO

Command

java

Arguments

-jar target/MCPServer-0.0.1-SNAPSHOT

> Environment Variables

Server EntryServers File

> Authentication

> Configuration

RestartDisconnect

Connected

nutriopt-mcp-server

Version: 0.0.1

Logging Level

debug

ResourcesPromptsToolsTasksAppsPingSamplingElicitationsRootsAuthMetadata

Tools

List Tools

Clear

ping

Simple ping test tool

History

Clear

4. tools/call

ping

Simple ping test tool

X Read-only ✓ Destructive X Idempotent ✓ Open-world

Tool-specific Metadata:

No metadata pairs.

Run Tool

Copy Input

Tool Result: Success

""pong""

Server Notifications

1. notifications/message

OpenPrices tool works

MCP Inspector v0.22.0

Transport Type

STDIO

Command

java

Arguments

-jar target/MCPServer-0.0.1-SNAPSHOT

> Environment Variables

Server EntryServers File

> Authentication

> Configuration

RestartDisconnect

Connected

nutriopt-mcp-server

Version: 0.0.1

ResourcesPromptsToolsTasksAppsPingSamplingElicitationsRootsAuthMetadata

Tools

List Tools

Clear

getOpenPricesByProductCode

Get product price records from Open Prices using a product barcode or product code.

ping

Simple ping test tool

getOpenPricesByProductCode

Get product price records from Open Prices using a product barcode or product code.

X Read-only ✓ Destructive X Idempotent ✓ Open-world

productCode *

quaker oats

Tool-specific Metadata:

No metadata pairs.

Add Pair

Run Tool

Copy Input

Tool Result: Success

""Open Prices results for product code quaker oats: {\\"items\\":[],\\"page\\":1,\\"pages\\":1,\\"size\\":10,\\"total\\":0}""

FoodStorageTool working

Tools

List Tools

Clear

getOpenPricesByProductCode

Get product price records from Open Prices using a product barcode or product code.

listSavedFoods

List all saved food items from MariaDB with their IDs. Use these IDs when calling the LP optimizer.

ping

Simple ping test tool

findSavedFoodsByName

Find saved food items in MariaDB by food name. Useful before saving a duplicate food item.

saveFoodItems

Save one or more food items with nutrition and price information into MariaDB. If a barcode already exists, update the existing food item. Returns saved food IDs.

saveFoodItems

Save one or more food items with nutrition and price information into MariaDB. If a barcode already exists, update the existing food item. Returns saved food IDs.

X Read-only

✓ Destructive

X Idempotent

✓ Open-world

requests *

Copy JSON

Format JSON

Switch to Fc

```
[
  {
    "name": "Quaker Large Flake Oats",
    "genericName": "oats",
    "brand": "Quaker",
    "barcode": "055577105026",
    "quantityText": "1 kg",
    "packageSizeGrams": 1000,
    "caloriesPer100g": 389,
    "proteinPer100g": 16.9,
    "carbsPer100g": 66.3,
    "fatPer100g": 6.9,
    "fiberPer100g": 10.6,
    "latestPrice": 4.99,
    "currency": "CAD",
    "costPer100g": 0.499,
    "priceSource": "manual_fallback",
    "priceDate": "2026-06-08",
    "minGrams": 30,
    "maxGrams": 120
  }
]
```

```
[
  {
    "name": "Quaker Large Flake Oats",
    "genericName": "oats",
    "brand": "Quaker",
    "barcode": "055577105026",
    "quantityText": "1 kg",
    "packageSizeGrams": 1000,
    "caloriesPer100g": 389,
    "proteinPer100g": 16.9,
    "carbsPer100g": 66.3,
    "fatPer100g": 6.9,
    "fiberPer100g": 10.6,
    "latestPrice": 4.99,
    "currency": "CAD",
    "costPer100g": 0.499,
    "priceSource": "manual_fallback",
    "priceDate": "2026-06-08",
    "minGrams": 30,
    "maxGrams": 120
  }
]
```

Tool-specific Metadata:

Add Pair

No metadata pairs.

 Run Tool

 Copy Input

Tool Result: **Success**

```
[
  {
    id: 1
    name: "Quaker Large Flake Oats"
    genericName: "oats"
    brand: "Quaker"
    barcode: "055577105026"
    caloriesPer100g: 389
    proteinPer100g: 16.9
    costPer100g: 0.499
    status: "SAVED"
  }
]
```



Now it shows LpOptimizerTool also

ResourcesPromptsToolsTasksAppsPingSamplingElicitationsRootsAuthMetadata

Tools

List Tools

Clear

optimizeDietLp

Optimize a one-day diet using Linear Programming. Reads selected foods from MariaDB by food IDs and minimizes daily cost while satisfying macro constraints.

getOpenPricesByProductCode

Get product price records from Open Prices using a product barcode or product code.

listSavedFoods

List all saved food items from MariaDB with their IDs. Use these IDs when calling the LP optimizer.

ping

Simple ping test tool

findSavedFoodsByName

Find saved food items in MariaDB by food name. Useful before saving a duplicate food item.

saveFoodItems

Select a tool

Select a tool from the list to view its details and run it

History

Clear

Server Notifications

```
[
  {
    "name": "Quaker Large Flake Oats",
    "genericName": "oats",
    "brand": "Quaker",
    "barcode": "055577105026",
    "quantityText": "1 kg",
    "packageSizeGrams": 1000.0,
    "caloriesPer100g": 389.0,
    "proteinPer100g": 16.9,
    "carbsPer100g": 66.3,
    "fatPer100g": 6.9,
    "fiberPer100g": 10.6,
    "latestPrice": 4.99,
    "currency": "CAD",
    "costPer100g": 0.499,
    "priceSource": "manual_test",
    "priceDate": "2026-06-08",
    "minGrams": 30.0,
    "maxGrams": 150.0
  },
  {
    "name": "Whole Milk 3.25%",
    "genericName": "milk",
    "brand": "Neilson",
    "barcode": "064420001234",
    "quantityText": "4 L",
    "packageSizeGrams": 4000.0,
    "caloriesPer100g": 61.0,
    "proteinPer100g": 3.2,
```

```
"carbsPer100g": 4.8,
"fatPer100g": 3.3,
"fiberPer100g": 0.0,
"latestPrice": 6.49,
"currency": "CAD",
"costPer100g": 0.162,
"priceSource": "manual_test",
"priceDate": "2026-06-08",
"minGrams": 200.0,
"maxGrams": 700.0
},
{
  "name": "Large Eggs",
  "genericName": "eggs",
  "brand": "No Name",
  "barcode": "060383000111",
  "quantityText": "12 eggs",
  "packageSizeGrams": 600.0,
  "caloriesPer100g": 143.0,
  "proteinPer100g": 12.6,
  "carbsPer100g": 0.7,
  "fatPer100g": 9.5,
  "fiberPer100g": 0.0,
  "latestPrice": 3.99,
  "currency": "CAD",
  "costPer100g": 0.665,
  "priceSource": "manual_test",
  "priceDate": "2026-06-08",
  "minGrams": 50.0,
  "maxGrams": 200.0
},
{
  "name": "Chicken Breast",
  "genericName": "chicken breast",
  "brand": "Generic",
  "barcode": "000000000101",
  "quantityText": "1 kg",
  "packageSizeGrams": 1000.0,
  "caloriesPer100g": 165.0,
  "proteinPer100g": 31.0,
  "carbsPer100g": 0.0,
  "fatPer100g": 3.6,
  "fiberPer100g": 0.0,
  "latestPrice": 12.00,
  "currency": "CAD",
  "costPer100g": 1.2,
  "priceSource": "manual_test",
  "priceDate": "2026-06-08",
  "minGrams": 100.0,
```

```
"maxGrams": 350.0
},
{
  "name": "White Rice Cooked",
  "genericName": "rice",
  "brand": "Generic",
  "barcode": "000000000102",
  "quantityText": "cooked",
  "packageSizeGrams": 1000.0,
  "caloriesPer100g": 130.0,
  "proteinPer100g": 2.7,
  "carbsPer100g": 28.0,
  "fatPer100g": 0.3,
  "fiberPer100g": 0.4,
  "latestPrice": 2.50,
  "currency": "CAD",
  "costPer100g": 0.25,
  "priceSource": "manual_test",
  "priceDate": "2026-06-08",
  "minGrams": 100.0,
  "maxGrams": 500.0
},
{
  "name": "Banana",
  "genericName": "banana",
  "brand": "Generic",
  "barcode": "000000000103",
  "quantityText": "1 kg",
  "packageSizeGrams": 1000.0,
  "caloriesPer100g": 89.0,
  "proteinPer100g": 1.1,
  "carbsPer100g": 22.8,
  "fatPer100g": 0.3,
  "fiberPer100g": 2.6,
  "latestPrice": 1.69,
  "currency": "CAD",
  "costPer100g": 0.169,
  "priceSource": "manual_test",
  "priceDate": "2026-06-08",
  "minGrams": 100.0,
  "maxGrams": 300.0
},
{
  "name": "Peanut Butter",
  "genericName": "peanut butter",
  "brand": "Kraft",
  "barcode": "068100091234",
  "quantityText": "1 kg",
  "packageSizeGrams": 1000.0,
```

```

    "caloriesPer100g": 588.0,
    "proteinPer100g": 25.0,
    "carbsPer100g": 20.0,
    "fatPer100g": 50.0,
    "fiberPer100g": 6.0,
    "latestPrice": 5.99,
    "currency": "CAD",
    "costPer100g": 0.599,
    "priceSource": "manual_test",
    "priceDate": "2026-06-08",
    "minGrams": 10.0,
    "maxGrams": 60.0
  },
  {
    "name": "Frozen Mixed Vegetables",
    "genericName": "mixed vegetables",
    "brand": "No Name",
    "barcode": "060383001234",
    "quantityText": "750 g",
    "packageSizeGrams": 750.0,
    "caloriesPer100g": 65.0,
    "proteinPer100g": 3.0,
    "carbsPer100g": 12.0,
    "fatPer100g": 0.5,
    "fiberPer100g": 4.0,
    "latestPrice": 2.99,
    "currency": "CAD",
    "costPer100g": 0.399,
    "priceSource": "manual_test",
    "priceDate": "2026-06-08",
    "minGrams": 100.0,
    "maxGrams": 400.0
  }
]

```

Response

```

[
  {
    "id": 1,
    "name": "Quaker Large Flake Oats",
    "genericName": "oats",
    "brand": "Quaker",
    "barcode": "055577105026",
    "caloriesPer100g": 389,
    "proteinPer100g": 16.9,
    "costPer100g": 0.499,
    "status": "UPDATED"
  },

```

```
{
  "id": 2,
  "name": "Whole Milk 3.25%",
  "genericName": "milk",
  "brand": "Neilson",
  "barcode": "064420001234",
  "caloriesPer100g": 61,
  "proteinPer100g": 3.2,
  "costPer100g": 0.162,
  "status": "SAVED"
},
{
  "id": 3,
  "name": "Large Eggs",
  "genericName": "eggs",
  "brand": "No Name",
  "barcode": "060383000111",
  "caloriesPer100g": 143,
  "proteinPer100g": 12.6,
  "costPer100g": 0.665,
  "status": "SAVED"
},
{
  "id": 4,
  "name": "Chicken Breast",
  "genericName": "chicken breast",
  "brand": "Generic",
  "barcode": "000000000101",
  "caloriesPer100g": 165,
  "proteinPer100g": 31,
  "costPer100g": 1.2,
  "status": "SAVED"
},
{
  "id": 5,
  "name": "White Rice Cooked",
  "genericName": "rice",
  "brand": "Generic",
  "barcode": "000000000102",
  "caloriesPer100g": 130,
  "proteinPer100g": 2.7,
  "costPer100g": 0.25,
  "status": "SAVED"
},
{
  "id": 6,
  "name": "Banana",
  "genericName": "banana",
  "brand": "Generic",
```

```

    "barcode": "000000000103",
    "caloriesPer100g": 89,
    "proteinPer100g": 1.1,
    "costPer100g": 0.169,
    "status": "SAVED"
  },
  {
    "id": 7,
    "name": "Peanut Butter",
    "genericName": "peanut butter",
    "brand": "Kraft",
    "barcode": "068100091234",
    "caloriesPer100g": 588,
    "proteinPer100g": 25,
    "costPer100g": 0.599,
    "status": "SAVED"
  },
  {
    "id": 8,
    "name": "Frozen Mixed Vegetables",
    "genericName": "mixed vegetables",
    "brand": "No Name",
    "barcode": "060383001234",
    "caloriesPer100g": 65,
    "proteinPer100g": 3,
    "costPer100g": 0.399,
    "status": "SAVED"
  }
]

```

Saved into MariaDB

```

MariaDB [(none)]> use nutriopt_db;
Database changed
MariaDB [nutriopt_db]> show tables;
+-----+
| Tables_in_nutriopt_db |
+-----+
| food_items             |
+-----+
1 row in set (0.005 sec)

MariaDB [nutriopt_db]> select * from food_items;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | barcode | brand | calories_per100g | carbs_per100g | cost_per100g | currency | fat_per100g | fiber_per100g | generic_name | latest_price | max_grams | min_grams | name | package_size_grams | price_date | price_source | protein_per100g | quantity_text |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | 055577105026 | Quaker | 389 | 66.3 | 0.499 | CAD | 6.9 | 10.6 | oats | 4.99 | 120 | 30 | Quaker Large Flake Oats | 1000 | 2026-06-08 | manual_fallback | 16.9 | 1 kg |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.011 sec)

MariaDB [nutriopt_db]>

```

```
MariaDB [nutriopt_db]> select * from food_items;
```

id	barcode	brand	calories_per100g	carbs_per100g	cost_per100g	currency	fat_per100g	fiber_per100g	generic_name	latest_price
max_grams	min_grams	name	package_size_grams	price_date	price_source	protein_per100g	quantity_text			
1	055577105026	Quaker	389	66.3	0.499	CAD	6.9	10.6	oats	4.99
150	30	Quaker Large Flake Oats	1000	2026-06-08	manual_test		16.9	1 kg		
2	064420001234	Neilson	61	4.8	0.162	CAD	3.3	0	milk	6.49
700	200	Whole Milk 3.25%	4000	2026-06-08	manual_test		3.2	4 L		
3	060383000111	No Name	143	0.7	0.665	CAD	9.5	0	eggs	3.99
200	50	Large Eggs	600	2026-06-08	manual_test		12.6	12 eggs		
4	000000000101	Generic	165	0	1.2	CAD	3.6	0	chicken breast	12
350	100	Chicken Breast	1000	2026-06-08	manual_test		31	1 kg		
5	000000000102	Generic	130	28	0.25	CAD	0.3	0.4	rice	2.5
500	100	White Rice Cooked	1000	2026-06-08	manual_test		2.7	cooked		
6	000000000103	Generic	89	22.8	0.169	CAD	0.3	2.6	banana	1.69
300	100	Banana	1000	2026-06-08	manual_test		1.1	1 kg		
7	068100091234	Kraft	588	20	0.599	CAD	50	6	peanut butter	5.99
60	10	Peanut Butter	1000	2026-06-08	manual_test		25	1 kg		
8	060383001234	No Name	65	12	0.399	CAD	0.5	4	mixed vegetables	2.99
400	100	Frozen Mixed Vegetables	750	2026-06-08	manual_test		3	750 g		

optimizeDietLp Payload

```
{
  "foodIds": [1, 2, 3, 4, 5, 6, 7, 8],
  "targets": {
    "budgetMax": 20.0,
    "caloriesMax": 3000.0,
    "caloriesMin": 2000.0,
    "carbsMax": 450.0,
    "carbsMin": 180.0,
    "fatMax": 130.0,
    "fatMin": 40.0,
    "fiberMin": 15.0,
    "proteinMin": 100.0
  }
}
```

optimizeDietLp

```
{
  "foodIds": [
    1,
    2,
    3,
    4,
    5,
    6,
    7,
    8
  ],
  "targets": {
    "budgetMax": 20,
    "caloriesMax": 3000,
    "caloriesMin": 2000,
    "carbsMax": 450,
    "carbsMin": 180,
```

```
"fatMax": 130,  
"fatMin": 40,  
"fiberMin": 15,  
"proteinMin": 100  
}  
}
```

```
{  
  "foodIds": [  
    1,  
    2,  
    3,  
    4,  
    5,  
    6,  
    7,  
    8  
  ],  
  "targets": {  
    "budgetMax": 20,  
    "caloriesMax": 3000,  
    "caloriesMin": 2000,  
    "carbsMax": 450,  
    "carbsMin": 180,  
    "fatMax": 130,  
    "fatMin": 40,  
    "fiberMin": 15,  
    "proteinMin": 100  
  }  
}
```

Response

```
{  
  "status": "OPTIMAL",  
  "totalCalories": 2000,  
  "totalProtein": 100,  
  "totalCarbs": 273.19,  
  "totalFat": 58.03,  
  "totalFiber": 27.76,  
  "totalCost": 4.6,  
  "items": [  
    {  
      "foodId": 1,  
      "food": "Quaker Large Flake Oats",  
      "quantity": 150,  
      "unit": "g/ml"  
    },  
  ],  
}
```

```
{
  "foodId": 2,
  "food": "Whole Milk 3.25%",
  "quantity": 220.8,
  "unit": "g/ml"
},
{
  "foodId": 3,
  "food": "Large Eggs",
  "quantity": 50,
  "unit": "g/ml"
},
{
  "foodId": 4,
  "food": "Chicken Breast",
  "quantity": 100,
  "unit": "g/ml"
},
{
  "foodId": 5,
  "food": "White Rice Cooked",
  "quantity": 414.24,
  "unit": "g/ml"
},
{
  "foodId": 6,
  "food": "Banana",
  "quantity": 100,
  "unit": "g/ml"
},
{
  "foodId": 7,
  "food": "Peanut Butter",
  "quantity": 60,
  "unit": "g/ml"
},
{
  "foodId": 8,
  "food": "Frozen Mixed Vegetables",
  "quantity": 100,
  "unit": "g/ml"
}
]
```

Tool Result: **Success**

```
{
  status: "OPTIMAL"
  totalCalories: 2000
  totalProtein: 100
  totalCarbs: 273.19
  totalFat: 58.03
  totalFiber: 27.76
  totalCost: 4.6
  items: [
    0: {
      foodId: 1
      food: "Quaker Large Flake Oats"
      quantity: 150
      unit: "g/ml"
    }
    1: {
      foodId: 2
      food: "Whole Milk 3.25%"
      quantity: 220.8
      unit: "g/ml"
    }
  ]
}
```



